Prior-adaptive evidence thresholds for ACMG/AMP variant classification

Abstract

Clinical variant classification under the ACMG/AMP guidelines converts a piece of evidence (for example, a functional assay result) into an evidence strength (Supporting, Moderate, Strong, Very Strong) by comparing it against a threshold. The most widely used thresholds, from Tavtigian and colleagues, were derived using a single reference prior probability of pathogenicity ($p = 0.10$) and are then applied unchanged to genes with very different priors. We show that this produces systematically wrong evidence assignments away from that one reference point, especially for benign evidence at low priors, where the standard thresholds are far too strict and quietly discard real benign signal. We describe an alternative, ACMG-Bayes, that recomputes the correct threshold for the gene's actual prior rather than reusing a single fixed value, and prove that this is the only way to get exact results at every prior. We further show that even the standard posterior targets used to define "Likely Benign" and "Benign" (10% and 1%) are themselves poorly matched to a low prior: at the reference prior of 0.10, the Likely Benign target requires no evidence at all to trigger. Shifting these targets to 1% and 0.1% is better justified and, on 89 real calibrated functional assay datasets, nearly doubles pathogenic sensitivity (89.8%→97.0%) and reduces dangerous false-negative benign calls by 47%, at matched overall coverage and comparable specificity.

Introduction

The ACMG/AMP guidelines [1] classify a genetic variant into one of five categories based on how likely it is to be disease-causing: Pathogenic ($\geq$99% probability), Likely Pathogenic (LP; $\geq$90%), Uncertain Significance (VUS; in between), Likely Benign (LB; $\leq$10%), and Benign ($\leq$1%). Reaching one of these categories requires combining independent pieces of evidence – population data, computational predictions, functional assays, segregation in families, and so on – each of which is assigned a strength (Supporting, Moderate, Strong, or Very Strong, in either the pathogenic or benign direction) according to fixed combination rules [1].

Tavtigian and colleagues [2, 3] showed that this scheme can be put on a rigorous statistical footing: each evidence strength corresponds to a specific likelihood ratio (how many times more likely the evidence is if the variant is pathogenic than if it is benign), and the strengths were assigned point values ($\pm1, \pm2, \pm4, \pm8$) so that adding points together corresponds to multiplying likelihood ratios. The point-to-likelihood-ratio conversion, however, was fixed once, using a single reference gene-level prior probability of pathogenicity, $p = 0.10$. Modern calibration pipelines – including our own ExCALIBR [7] – estimate a different prior for every gene from population frequency data, and apply the same, unchanged Tavtigian thresholds regardless of what that prior turns out to be.

This matters because the correct evidence threshold genuinely depends on the prior: the same functional assay result that reaches Likely Pathogenic at $p = 0.10$ would leave a variant squarely in the uncertain range at $p = 0.01$ and would be far too weak a bar at $p = 0.25$. Harrison and Rehm [6] observed exactly this kind of problem in practice: variants classified Likely Pathogenic in

Table 1: What posterior probability a variant actually has when it just reaches each ACMG boundary, standard (Tavtigian) thresholds vs. the intended target, at five representative priors. "AB" columns are ACMG-Bayes, which is exact at every target by construction (see Results below and Methods).

p	C^*	LP (target 0.90)		P (target 0.99)		LB (target 0.10)		B (target 0.01)	
		Tav	AB	Tav	AB	Tav	AB	Tav	AB
0.01	8511	**.900**	.900	.999	**.990**	.003	**.100**	.000	**.010**
0.05	950	**.900**	.900	.996	**.990**	.022	**.100**	.000	**.010**
0.10	350	**.900**	.900	.994	**.990**	.051	**.100**	.001	**.010**
0.25	92	.908	**.900**	**.990**	.990	.159	**.100**	.006	**.010**
0.45	2876	.997	**.900**	1.00	**.990**	.232	**.100**	.001	**.010**

ClinVar often turn out, on reanalysis, to carry nowhere near the intended 90% confidence. Reusing a single set of thresholds across genes with different priors is a plausible mechanistic explanation for that inconsistency.

In this paper we (1) quantify how badly the standard, prior-fixed thresholds miscalibrate evidence assignments away from $p = 0.10$; (2) show, with a short mathematical argument (details in Methods), that no scheme that adds fixed integer points together can give exact results at every prior – some form of prior-adaptation is unavoidable; (3) describe ACMG-Bayes, a straightforward fix that recomputes the correct likelihood-ratio threshold for each gene's own prior, while still producing an ACMG-familiar point label for reporting; (4) show that the standard Likely-Benign/Benign posterior targets are themselves poorly chosen relative to a typical prior, and propose a better-justified alternative; and (5) validate that alternative on 89 real calibrated functional assay datasets.

Results

Standard thresholds are only correct at the one prior they were calibrated to

Table 1 shows, at five representative priors, what posterior probability a variant actually has when it just reaches each of the four ACMG boundaries under the standard (Tavtigian) thresholds, compared to the boundary the ACMG guidelines intend (90%, 99%, 10%, 1%). At the reference prior, $p = 0.10$, the two agree closely, by construction. Away from it, they diverge substantially, and the pathogenic and benign directions fail differently.

The pathogenic (LP/P) boundaries stay reasonably close to their intended targets for priors between roughly 0.01 and 0.25. Above that, the standard approach becomes over-confident: at $p = 0.45$ a variant reaching the LP boundary actually has a 99.7% posterior, not 90%. The benign (LB) boundary fails everywhere shown except by coincidence near $p = 0.25$: at $p = 0.01$, a variant that just reaches the LB boundary under the standard thresholds actually has only a 0.3% chance of being pathogenic – far below the intended 10% cutoff. In practical terms, at $p = 0.01$ the standard likelihood-ratio threshold for Likely Benign evidence is 42-fold stricter than it needs to be to hit the intended 10% target (a likelihood ratio below 0.26 is required, versus the 11 that would actually suffice). Every variant whose evidence falls in that gap is being denied benign evidence it has genuinely earned. Figures 1 and 2 show this across the full prior range.

No fixed set of thresholds can be exact at every prior

This is not a matter of picking a better reference prior. We show formally (Methods) that any classification scheme built from fixed evidence-strength tiers (Supporting/Moderate/Strong/Very Strong, in a fixed 1:2:4:8 point ratio) that are simply added together, using a single fixed set of combination thresholds, *cannot* give exact posterior probabilities at every prior – it can be exact at one prior, or a small number of them, but not all. Something has to give: either the thresholds have to move with the prior, or the points have to stop being the thing that gets added together. The Tavtigian system keeps fixed thresholds and fixed additive points, at the cost of exactness (Table 1). ACMG-Bayes, described next, keeps fixed evidence combination rules and exact posteriors, at the cost of letting the point-to-evidence conversion vary with the prior – which turns out to be the only part of the system that needs to.

ACMG-Bayes: recompute the threshold for the gene's actual prior

ACMG-Bayes addresses the problem directly: instead of reusing one likelihood-ratio threshold for every gene, it computes, from Bayes' theorem, the exact likelihood ratio needed to reach each ACMG boundary (90%, 99%, 10%, 1%) at whatever prior the gene actually has (derivation in Methods). This guarantees the four boundary posteriors are correct by construction, at every prior – the "AB" columns in Table 1 are exact, not approximate. Figure 3 shows the geometric picture: where the standard approach is a single straight line that only touches all four correct answers when $p = 0.10$, ACMG-Bayes is a line that shifts to always pass through the four correct answers, whatever the prior is.

When more than one piece of evidence needs to be combined, ACMG-Bayes combines the underlying likelihood ratios directly (equivalently, adds their logarithms) rather than rounding each piece of evidence to an integer point value first and adding those. This is exact Bayesian evidence combination regardless of prior, and it is the only way to combine evidence that is consistent with also having exact, prior-specific thresholds. A final combined value can still be converted back into a familiar ACMG point label for reporting (e.g. "moderate-equivalent") – but that label is computed *after* combination, for display only, and its exact numeric meaning is specific to the gene's own prior, so it cannot be read off a single table shared across genes with different priors (Methods).

The standard Likely Benign and Benign targets are themselves miscalibrated

Table 1 shows what happens when the standard 90%/99%/ 10%/1% *targets* are pursued using the wrong threshold. A separate question is whether those four targets are even well-chosen relative to a realistic prior, independent of any threshold-calibration issue.

They are not, for the two benign-direction targets. The four ACMG targets (1%, 10%, 90%, 99%) are symmetric around 50-50 odds: LP and LB are each "10 points from the extreme," P and B are each "1 point from the extreme." That symmetry is only evidence-neutral if the reference prior is itself 50%. It is not – the reference prior used throughout this paper (and in the original Tavtigian calibration) is 10%. At exactly that prior, the amount of evidence required to reach the Likely Benign target works out to *zero*: a likelihood ratio of 1, i.e. no data at all, is already enough (derivation in Methods, Table 2). More generally, for any gene with a prior below 10% – which is most of them – the standard Likely Benign target requires either no evidence or, worse, evidence that actually leans pathogenic. The Likely Pathogenic and Pathogenic targets are not affected the same way; the problem is specific to the benign-direction targets.

A better-justified pair of targets is LB= 1% and B= 0.1% – an order of magnitude stricter than the current standard, and, unlike the current standard, one that actually requires real benign-leaning

Table 2: Likelihood ratio required to reach each target, at the reference prior $p = 0.10$. A likelihood ratio of exactly 1 means the target is reached with no evidence at all; a ratio below 1 means genuinely benign-leaning evidence is required; a ratio above 1 means pathogenic-leaning evidence would (wrongly) be required.

Tier	Target posterior	Likelihood ratio required
LP (unchanged)	0.90	81.0
LB, standard ACMG	0.10	1.00 *(no evidence needed)*
LB, shifted target (evaluated below)	0.01	0.091
P (unchanged)	0.99	891
B, standard ACMG	0.01	0.091
B, shifted target (evaluated below)	0.001	0.0090

evidence to trigger. (A value that mirrors LP/P exactly in the amount of evidence required would be closer to 0.14%/0.01%; 1%/0.1% is a practical, round-number compromise that is already a large improvement – Methods.)

Validation on 89 real calibrated functional assay datasets

We tested the shifted targets (LB= 1%, B= 0.1%) against the standard targets on 89 real deep mutational scanning / multiplexed assay of variant effect datasets, using the same underlying assay calibrations and the same gene-specific priors for both, so that any difference reflects the evidence-threshold choice alone. Ground truth came from each variant's own clinical curation label (Pathogenic/Likely Pathogenic or Benign/Likely Benign); variants without one of those two labels were excluded from the comparison, leaving 23,717 control variants (9,441 pathogenic, 14,276 benign) across 72 evaluated dataset components.

Table 3: How true pathogenic and benign controls were classified (LP-or-P vs. VUS/indeterminate vs. LB-or-B), standard targets vs. shifted targets (LB= 1%, B= 0.1%), same 89 datasets and priors.

Standard targets (LB=10%, B=1%)			
True control	LP or P	VUS	LB or B
Pathogenic ($n = 9{,}441$)	2,049	7,160	232
Benign ($n = 14{,}276$)	15	3,591	10,670
Shifted targets (LB=1%, B=0.1%)			
True control	LP or P	VUS	LB or B
Pathogenic ($n = 9{,}441$)	3,885	5,434	122
Benign ($n = 14{,}276$)	40	6,025	8,211

At essentially matched overall coverage, shifting the LB/B targets nearly doubles pathogenic sensitivity (89.8%→97.0%) and cuts dangerous false negatives – real pathogenic variants wrongly called benign – by 47% (232→122), while specificity drops only marginally (99.86%→99.52%, both very high) and the dangerous false-positive count, while roughly tripling in relative terms, stays small in absolute terms (40 of 14,276 true benign controls, 0.28%). This matches what Table 2 predicts: the standard Likely Benign target gives away a benign call for evidence ranging from neutral all the

Table 4: Derived classification metrics from Table 3 (determinate calls only, i.e. excluding VUS).

Metric	Standard targets	Shifted targets
Coverage (determinate)	54.7%	51.7%
Sensitivity (path. controls resolved LP/P)	89.8%	**97.0%**
Specificity (benign controls resolved LB/B)	**99.86%**	99.52%
Accuracy	98.1%	**98.7%**
MCC	0.933	**0.970**
Dangerous false negatives (path. called LB/B)	232 (2.46%)	**122 (1.29%)**
Dangerous false positives (benign called LP/P)	**15 (0.11%)**	40 (0.28%)

way down to only mildly benign-leaning, and pathogenic controls are disproportionately the ones whose evidence happens to fall in that zone.

Discussion

Two independent sources of error compound in current assay calibration pipelines that use fixed, Tavtigian-style thresholds. First, the threshold itself is only correct at the one prior it was calibrated to, and drifts – sometimes by orders of magnitude in likelihood-ratio terms – at every other prior, with the largest practical impact on benign evidence at low priors, which is precisely the regime most relevant to rare disease genes. Second, independent of that, the standard Likely Benign and Benign targets themselves are calibrated as if the reference prior were 50%, when it is actually 10%, making the Likely Benign bar in particular too easy to clear.

ACMG-Bayes resolves the first problem by computing the correct threshold directly from the gene's own prior rather than reusing a fixed constant, and makes the second problem a simple configuration choice rather than a structural limitation: because thresholds are derived from targets rather than baked into a fixed point table, adopting better-justified targets (for example, 1%/0.1% instead of 10%/1% for LB/B) requires no change to the underlying method.

For calibration pipelines such as ExCALIBR [7], OddsPath [4], and PP3/BP4 calibration [5] that already estimate a gene-specific prior, the practical recommendation is straightforward: retain the underlying likelihood ratio (or its logarithm) as the working representation of evidence, convert to an ACMG-familiar point label only at the final reporting step, and derive the conversion thresholds from the gene's own prior rather than from a single shared table. This directly addresses the kind of miscalibration reported by Harrison and Rehm [6], where Likely Pathogenic calls in ClinVar often carry less confidence than the label implies.

A caveat is worth stating plainly: because the point-to-evidence conversion is prior-specific under ACMG-Bayes, a displayed point value (e.g. "Moderate-equivalent") is not directly comparable across two genes with different priors, even if the underlying evidence is numerically identical. This is intentional – it is the same underlying fact that makes fixed thresholds wrong in the first place – but it does mean point labels must be recomputed per gene rather than read off one shared table.

Methods

The Tavtigian point system

Given a prior p and a combined likelihood ratio $\mathrm{LR}^+ = C^{T/8}$ for total evidence points T, the posterior probability of pathogenicity is

$$P(Y{=}1 \mid E{=}e) = \frac{C^{T/8}\,p}{(C^{T/8}-1)p+1}. \tag{1}$$

The constant C (the "OddsPath" O_{PVSt}) is determined by the 2020 derivation [3]: five of the six LP combination rules each require odds equivalent to six pieces of Supporting evidence, fixing $O_{\mathrm{PSu}}^6 = 81$, hence $C^* = O_{\mathrm{PSu}}^8 = 81^{4/3} \approx 350$ at $p = 0.10$. We follow the modification of Tavtigian et al. [2] that resolves two originally inconsistent ACMG rules by moving the combination 2S to the LP classification (8 points, Post_P ≈ 0.975) and VS+M to the P classification (10 points, Post_P ≈ 0.994); all remaining rules are internally consistent at $p = 0.10$ with $C^* = 350$. Table 1 was produced by re-deriving C^* at each listed prior by the same grid search and evaluating Equation 1 at each ACMG boundary's point total ($T \in \{6, 10, -1, -7\}$).

Proof that no fixed, additive point system is exact at every prior

Theorem 1. *No scoring system satisfying fixed combination thresholds* (R1)*, the* 1:2:4:8 *tier ratio* (R2)*, and additive evidence combination* (R3) *can simultaneously achieve exact classification boundary posteriors at all priors* (R4)*.*

Proof. R2–R3 require $\log(\mathrm{LR}^+(T)) = \alpha T$ for a prior-independent constant α (i.e., a linear relationship between $\log(\mathrm{LR}^+)$ and T). R4 at the LP boundary then requires $e^{\alpha T_{LP}} = 9(1-p)/p$, which is strictly decreasing in p while the left side is constant. The equation holds for at most one p, contradicting R4. □

Corollary 1. *The same argument applies to all four classification boundaries, and any rescaling of the tier weights preserves the contradiction.*

Table 5 lays out the trade-off: Tavtigian satisfies R1–R3 at the cost of R4 (posteriors drift with prior, Table 1). A hypothetical "floating threshold" scheme could satisfy R3–R4 but breaks R1, since identical evidence combinations would then cross different classification boundaries at different priors – unworkable when other, non-prior-adaptive evidence sources (segregation, population data) are combined alongside it. ACMG-Bayes escapes the trade-off by relaxing R2 from a combination constraint to a *display-only* one: it never sums integer tiers – evidence is combined by summing $\log(\mathrm{LR}^+)$ directly – but any combined $\log(\mathrm{LR}^+)$ value can still be converted back into a Su/M/S/VS-equivalent label at the prior in use. Because R2 no longer constrains the combination arithmetic, Theorem 1's proof (which requires R2 and R3 together) does not apply, and R1, R3, and R4 all hold exactly and simultaneously.

ACMG-Bayes construction

At each prior p, the four Bayesian target LR^+ values are computed analytically [8]:

$${\mathrm{LR}^+}^*_q(p) = \frac{q(1-p)}{p(1-q)}, \qquad q \in \{0.99,\, 0.90,\, 0.10,\, 0.01\}, \tag{2}$$

Table 5: Any two of R1, R3, R4 can be satisfied exactly, but not all three under R2 (integer, fixed-ratio tiers used as the combination operand). ACMG-Bayes escapes this by relaxing R2 to a display-only constraint rather than dropping it outright.

Method	R1 (fixed thresh.)	R2 (1:2:4:8)	R3 (additive)	R4 (exact post.)
Tavtigian [2]	✓	✓	✓	×
Floating threshold	×	✓	✓	✓
ACMG-Bayes	✓	≈†	✓	✓

†R2 holds as a derived *display* label (Su/M/S/VS-equivalent, computed from the combined $\log(\mathrm{LR}^+)$ at the current prior), not as the combination operand.

yielding $P(Y{=}1 \mid E{=}e) = q$ exactly. These are anchored to the ACMG boundaries $T \in \{10, 6, -1, -7\}$ to define four knots of a piecewise-linear function $L_p(T) = \log(\mathrm{LR}^+(T))$, with linear interpolation within each of the three segments and extrapolation beyond the outermost knots using the adjacent segment's slope. The posterior at a single code (or point total) T is then $P(Y{=}1 \mid E{=}e) = \sigma\big(L_p(T) + \log \frac{p}{1-p}\big)$, where σ is the logistic function. By construction, $P(Y{=}1 \mid E{=}e) = q$ exactly at each of the four boundary values, for every prior $p \in (0, 1)$. The segment slopes ($\log(11)/6$, $\log(81)/7$, $\log(11)/4$ per point) are prior-independent; only the vertical position of the line shifts with prior (Figure 3).

When combining evidence from more than one code, ACMG-Bayes sums the per-code $\log(\mathrm{LR}^+)$ values directly (equivalently, multiplies the underlying likelihood ratios), which is exact Bayesian evidence aggregation regardless of prior: $\log(\mathrm{LR}^+_A \cdot \mathrm{LR}^+_B) = \log(\mathrm{LR}^+_A) + \log(\mathrm{LR}^+_B)$. Classification into P/LP/VUS/LB/B is then applied once, to the final combined $\log(\mathrm{LR}^+)$, using the same exact analytical thresholds derived above.

To recover an ACMG-familiar point label for reporting, the four boundary $\log(\mathrm{LR}^+)$ values at the gene's current prior are used to invert a combined $\log(\mathrm{LR}^+)$ value back into a (generally fractional) point count: locate which segment the value falls in, then linearly invert that segment's own slope to solve for T. Because the boundary $\log(\mathrm{LR}^+)$ values are prior-specific, the same numeric $\log(\mathrm{LR}^+)$ maps to a different point count at different priors – this is expected, not an error, and follows directly from Theorem 1.

ACMG-Bayes requires access to a continuous $\log(\mathrm{LR}^+)$ per evidence item, already available in pipelines like ExCALIBR that fit assay-score-to-LR^+ calibration curves directly [7]. For evidence sources only ever available as a discrete ACMG code (e.g. PM1, with no underlying continuous score), the code's canonical $\log(\mathrm{LR}^+)$ (its Tavtigian point value scaled by $\log(C^*)/8$ at $p = 0.10$, or the ACMG-Bayes-anchored equivalent at another prior) can be added into the same running total without approximation.

Evidence-symmetric correction to the LB/B targets

Writing $\mathrm{logit}(x) = \log\big(x/(1-x)\big)$, the exact evidence required to reach a target posterior q from prior p is

$$\log \mathrm{LR}^+_{\mathrm{req}}(q, p) = \mathrm{logit}(q) - \mathrm{logit}(p). \tag{3}$$

The standard ACMG targets satisfy $\mathrm{logit}(0.99) = +4.595$, $\mathrm{logit}(0.90) = +2.197$, $\mathrm{logit}(0.10) = -2.197$, $\mathrm{logit}(0.01) = -4.595$ – exactly symmetric about logit $= 0$, i.e. posterior $= 0.5$. By Equation 3, this design is evidence-neutral (LB and LP requiring equal and opposite amounts of evidence) only if $\mathrm{logit}(p) = 0$, i.e. the reference prior is itself 0.5. The reference prior used throughout this paper is

$p = 0.10$, giving, for the LB target,

$$\log \mathrm{LR}^{+}_{\mathrm{req}}(0.10,\ 0.10) = \mathrm{logit}(0.10) - \mathrm{logit}(0.10) = 0 \qquad (\mathrm{LR}^{+} = 1), \tag{4}$$

i.e. the standard LB target requires no evidence at all when the prior equals the reference prior. More generally, $\log \mathrm{LR}^{+\mathrm{LB}}_{\mathrm{req}}(p) = \mathrm{logit}(0.10) - \mathrm{logit}(p)$, which is positive (requiring pathogenic-leaning evidence for a benign call) whenever $p < 0.10$, zero at $p = 0.10$, and only correctly negative when $p > 0.10$. LP/P are not affected the same way: $\mathrm{logit}(0.90) - \mathrm{logit}(p)$ stays comfortably positive down to very low priors.

The evidence-symmetric correction mirrors LB/B to LP/P about the prior itself rather than about 50-50 odds:

$$\mathrm{logit}(q_{\mathrm{LB}}) = 2\,\mathrm{logit}(p) - \mathrm{logit}(q_{\mathrm{LP}}). \tag{5}$$

At $p = 0.10$, Equation 5 gives $q_{\mathrm{LB}} \approx 0.137\%$ (mirroring LP's 0.90) and, applying the same relation to $q_{\mathrm{P}} = 0.99$, $q_{\mathrm{B}} \approx 0.0125\%$ (mirroring P). The shifted targets evaluated in Results (LB= 1%, B= 0.1%) are a round-number compromise between the broken status quo and this exact value – an order of magnitude looser, but already sufficient to require genuine benign-leaning evidence rather than none (Table 2). As with C^{*} itself, this correction is stated at the reference prior; a fully general treatment would make the LB/B targets a function of the prior via Equation 5 rather than a second pair of fixed constants, which we do not pursue here.

Empirical validation

We compared the standard targets (LB=10%, B=1%) against the shifted targets (LB=1%, B=0.1%) using ExCALIBR bootstrap calibration fits for 89 real deep mutational scanning / multiplexed assay of variant effect datasets, applying identical assay-score-to-LR^{+} calibrations and identical gene-specific priors to both, so that any difference in outcome is attributable only to the evidence-threshold targets. Ground truth was taken from each variant's curated ClinVar sample label: a variant was treated as a pathogenic control if "Pathogenic/Likely Pathogenic" was one of its labels, a benign control if "Benign/Likely Benign" was one of its labels (synonymous-only variants were not counted as benign controls), and excluded from the comparison otherwise (e.g. plain population-frequency rows). Datasets and components were selected to match one evaluated component per dataset (preferring the 3-class over 2-class fit where both were available), giving 72 evaluated components and 23,717 total control variants. Classification metrics (sensitivity, specificity, accuracy, and Matthews correlation coefficient) were computed on the resulting 2×3 confusion matrix (true Pathogenic/Benign controls vs. predicted LP-or-P/VUS/LB-or-B), restricted to determinate (non-VUS) calls.

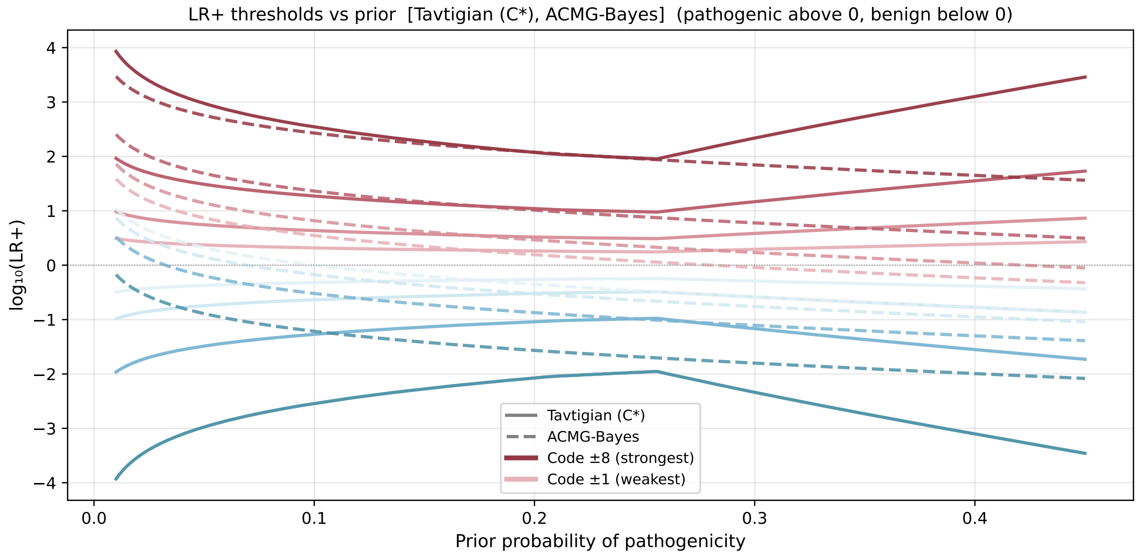

Figure 1: Likelihood-ratio thresholds for evidence codes 1, 2, 4, 8 (Su, M, S, VS) vs. gene-specific prior. Standard (Tavtigian, solid): follows a single fixed relationship; shows a non-monotone artefact near $p \approx 0.25$ and a large spike at $p = 0.45$ where benign combination rules drive C^* to 2876. ACMG-Bayes (dashed): varies smoothly with the prior and produces exact posteriors at each evidence boundary. Warm colours = pathogenic codes; cool colours = benign codes.

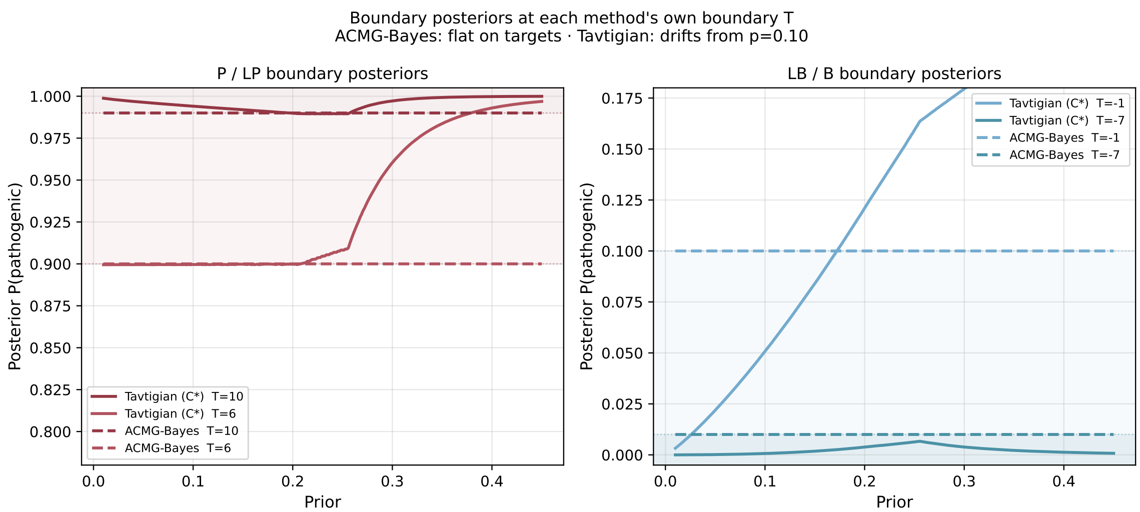

Figure 2: Posterior probability at each method's classification boundary vs. gene-specific prior. ACMG-Bayes (dashed): flat at all four targets by construction. Standard (Tavtigian, solid): well-calibrated for LP at priors ≤ 0.25 but severely miscalibrated for LB and B across the full prior range, with anomalous over-calibration near $p \approx 0.45$.

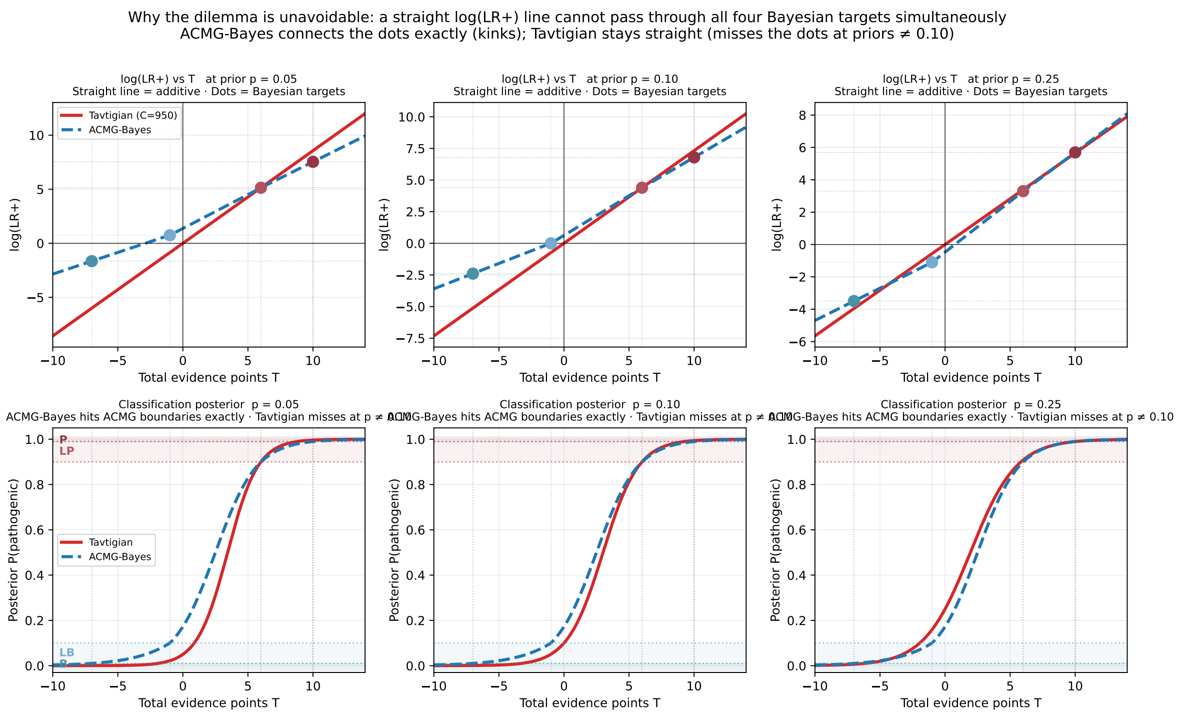

Figure 3: $\log(\mathrm{LR}^+)$ vs. total evidence points at three prior values. Coloured dots mark the exact Bayesian target at each of the four classification boundaries. Standard (Tavtigian, red): a single straight line that passes through all four dots only at $p = 0.10$. ACMG-Bayes (blue): a line that passes through all four dots at every prior by shifting vertically. Lower panels: the corresponding posterior probability curves.

Supplement: evidence tier point system and combination rules

Table 6: ACMG/AMP combination rules in the modified Tavtigian point system [2, 3]. Points: Su = 1, M = 2, S = 4, VS = 8; $Su_B = -1$, $S_B = -4$ (standard ACMG benign tiers only). †Modified rule: original ACMG classifies VS+M as LP and 2S as P; both are internally inconsistent at $p = 0.10$, so following Tavtigian et al. they are swapped. All calculations use the modified (swapped) rules.

Category	Combination	Points
P	VS + S	12
P	VS + 2M	12
P	VS + M + Su	11
P	VS + M †	10
P	S + 3M	10
P	S + 2M + 2Su	10
P	S + M + 4Su	10
LP	S + M	6
LP	S + 2Su	6
LP	3M	6
LP	2M + 2Su	6
LP	M + 4Su	6
LP	2S †	8
LB	$S_B + Su_B$	−5
LB	$2Su_B$	−2
B	$2S_B$	−8

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